

GREENTrace

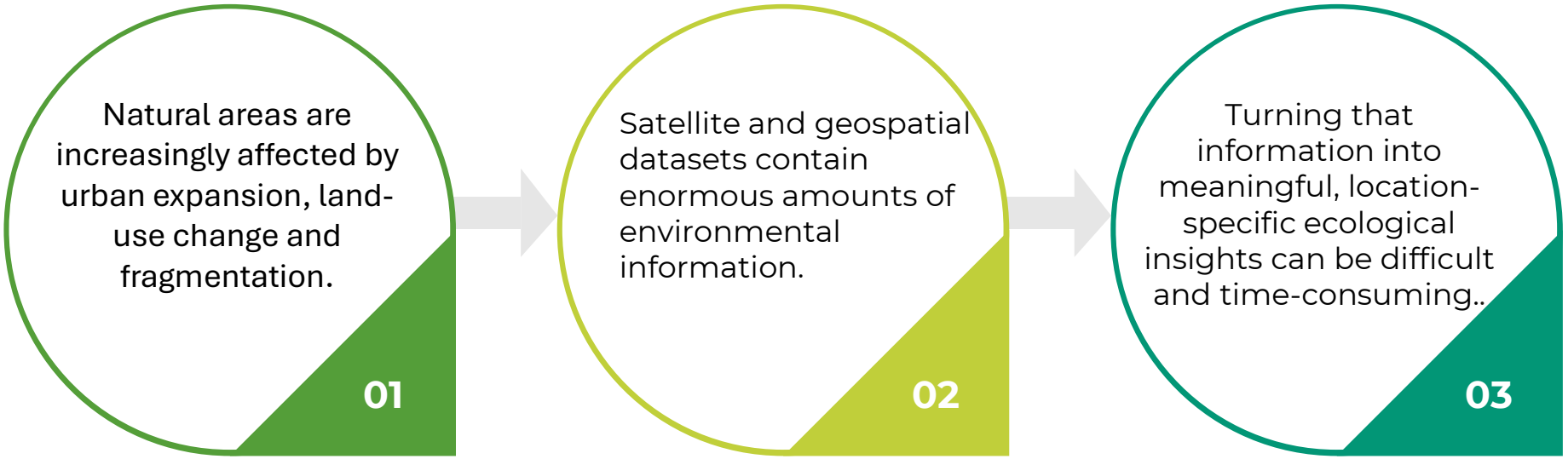
AI-Assisted Ecological Gap Analysis & Restoration Decision Support

By:

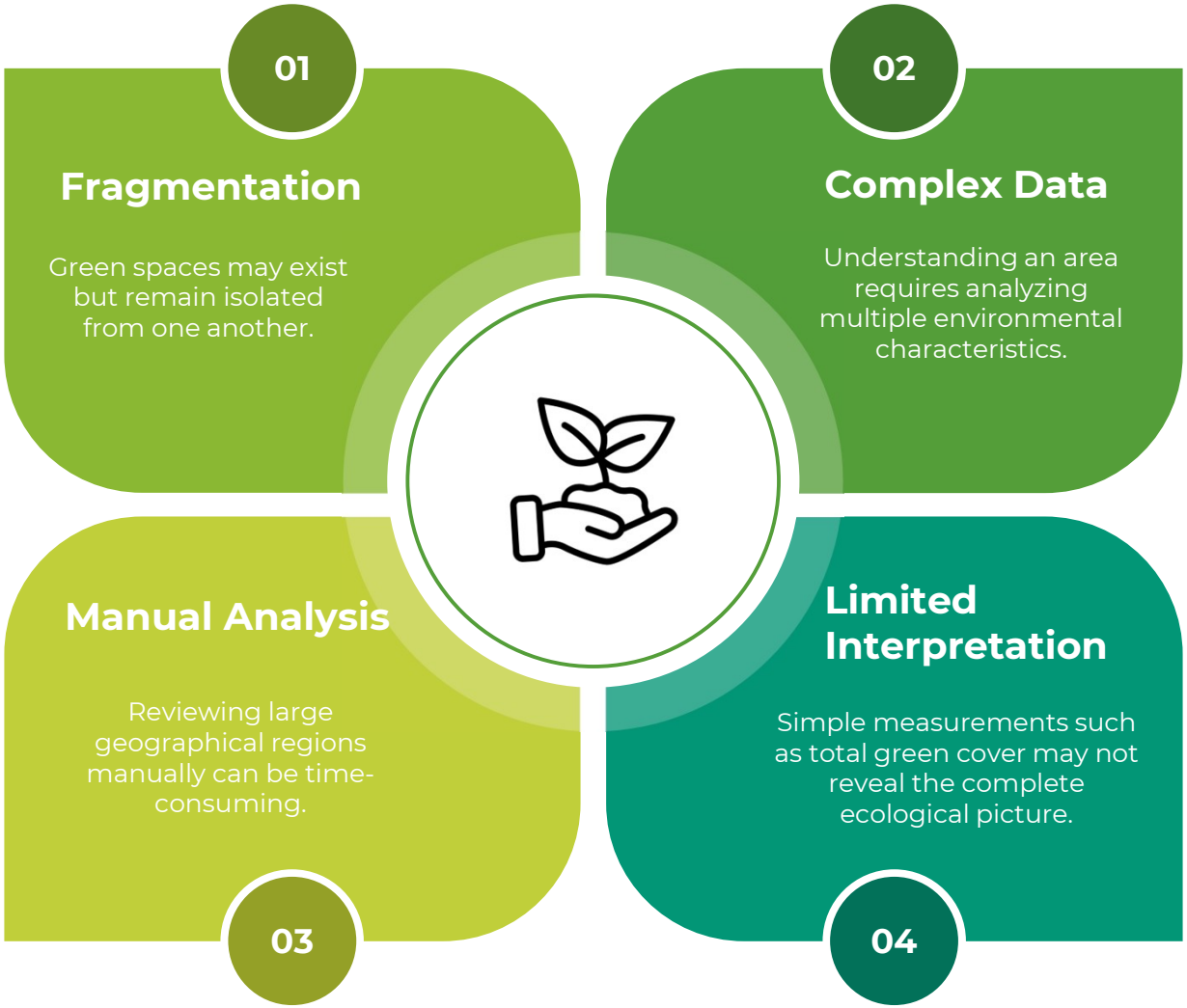
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Why GreenTrace?



The Problem


01
Fragmentation
Green spaces may exist but remain isolated from one another.
02
Complex Data
Understanding an area requires analyzing multiple environmental characteristics.
Manual Analysis
Reviewing large geographical regions manually can be time-consuming.
03
Limited Interpretation
Simple measurements such as total green cover may not reveal the complete ecological picture.
04

Introducing GREENTrace

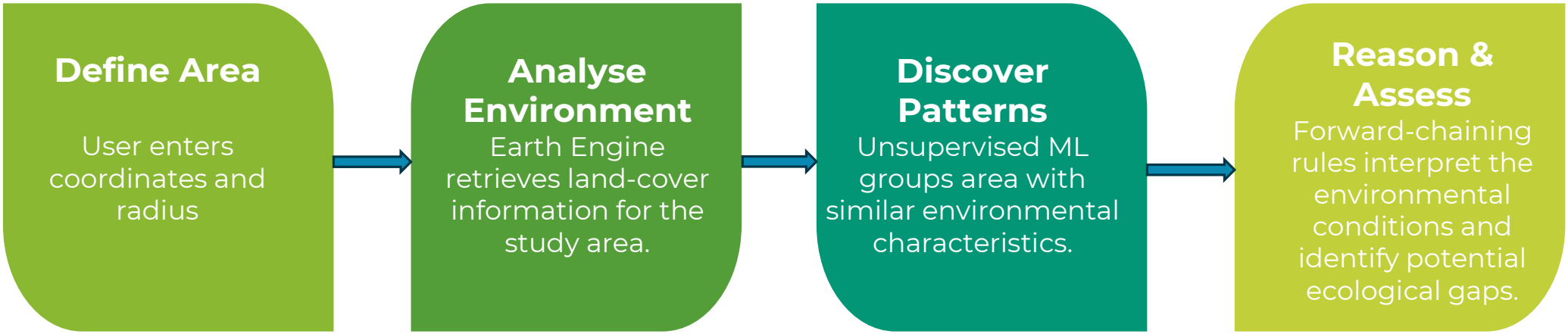
From location data to ecological insight

The user provides a geographical location using latitude and longitude, along with a radius defining the area to be analysed.



GreenTrace analyses the selected area and provides an environmental profile, identifies ecological patterns, and highlights potential areas for restoration consideration.

How GreenTrace Works



The system provides evidence and reasoning; the final decision remains with the user.

AI Techniques Used

Pattern discovery + logical reasoning = explainable ecological assessment



Technology & Data

Technology Stack

Streamlit will provide the interactive web interface through which users can enter a location and radius and view the results.

Scikit-learn will be used for the unsupervised machine-learning component.

Dynamic World will provide land-cover information such as trees, grass, water, crops, shrubland, and built-up areas.

Google Earth Engine will provide access to geospatial environmental data for the selected region.

Python will handle the environmental analysis, machine learning, and reasoning components.

Conclusion

GreenTrace analyses environmental characteristics within a user-defined geographical area.

It combines machine learning with knowledge-based reasoning to identify and interpret potential ecological gaps.

The system provides evidence for restoration decisions while keeping the final decision with humans.

Thank You